

The Complete Programming Commands Guide

Terminal • Python • Git • GitHub

1. Terminal Basics

Terminal / CLI

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| • <code>pwd</code> | Show the current path — the folder you're in right now |
| • <code>ls</code> / <code>dir</code> | List the contents of the current folder |
| • <code>cd</code> | Move into another folder |
| • <code>cd ..</code> | Go back one level, to the parent folder |
| • <code>cd ~</code> | Jump straight to your home folder |
| • <code>mkdir</code> | Create a new folder |
| • <code>touch</code> / <code>New-Item</code> | Create a new empty file |
| • <code>rm</code> / <code>Remove-Item</code> | Delete a file or folder |
| • <code>cp</code> / <code>Copy-Item</code> | Copy a file from one place to another
<code>cp original.txt backup.txt</code> |
| • <code>mv</code> / <code>ren</code> | Move or rename a file
<code>mv old.txt new.txt</code> |
| • <code>cat</code> | Print a file's contents directly in the terminal |
| • <code>nano</code> | Open a simple text editor inside the terminal |
| • <code>tree</code> | Show the folder and file structure as a tree |
| • <code>history</code> | Show a list of commands you've typed before |
| • <code>clear</code> / <code>cls</code> | Clear the terminal screen |

2. Python & Virtual Environments

Python & venv

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| • <code>python --version</code> | Check which Python version is installed on your machine |
| • <code>python -m venv venv</code> | Create a virtual environment to isolate the project's libraries from the rest of the machine |
| • <code>source venv/bin/activate</code> | Activate the virtual environment on Linux or macOS |
| • <code>venv\Scripts\activate.ps1</code> | Activate the virtual environment on Windows |
| • <code>deactivate</code> | Exit the virtual environment and return to the global setup |
| • <code>pip install package_name</code> | Install a Python library or package |
| • <code>pip uninstall package_name</code> | Remove an installed package |
| • <code>pip list</code> | Show all currently installed packages |
| • <code>pip freeze > requirements.txt</code> | Save the list of used packages to a file so they can be reinstalled later on any machine |
| • <code>pip install -r requirements.txt</code> | Install every package listed in the file at once |

3. Git Fundamentals

Git Fundamentals

Key Concepts

Working Directory

Staging Area

Local Repository

• <code>git --version</code>	Check which Git version is installed
• <code>git config --global user.name</code>	Set your name in Git's configuration <code>git config --global user.name "your_name"</code>
• <code>git config --global user.email</code>	Set your email in Git's configuration <code>git config --global user.email "you@example.com"</code>
• <code>git config --list</code>	Show all of Git's current configuration settings
• <code>git init</code>	Turn the current folder into a Git repository — the start of tracking a project
• <code>git status</code>	Check file status: modified, untracked, or ready to be saved
• <code>git add .</code>	Add all changes to the staging area, ready to be saved
• <code>git commit -m</code>	Save a snapshot of the changes with a short description <code>git commit -m "message"</code>
• <code>git log</code>	Show the history of all saved changes
• <code>git log --stat</code>	Show the change history along with details of files that changed
• <code>git restore</code>	Discard changes in a file that haven't been saved yet <code>git restore file.txt</code>
• <code>git restore --staged</code>	Unstage a file without losing the change itself <code>git restore --staged file.txt</code>
• <code>git reset</code>	Move the project's pointer to undo previously saved changes
• <code>git revert</code>	Safely undo a saved change without deleting the history

4. Branches & Merging

Branches & Merge

Key Concepts

Branch

Merge

Pull Request

Issues

Releases

Stars

Watch

• <code>git branch</code>	Show all branches that exist in the project
• <code>git switch -c</code>	Create a new branch and switch to it immediately <code>git switch -c branch_name</code>
• <code>git switch</code>	Switch to an existing branch <code>git switch branch_name</code>
• <code>git checkout</code>	The older way to switch branches, now replaced by git switch
• <code>git merge</code>	Merge a branch into the one you're currently on, e.g. merging a feature branch into main <code>git merge branch_name</code>

5. GitHub & Remote Repositories

GitHub & Remote Repositories

Key Concepts

Repository

Owner

Remote / origin

SSH Auth

• <code>ssh-keygen</code>	Generate an SSH key for secure authentication with GitHub
• <code>git clone</code>	Copy a repository from GitHub to your machine <code>git clone repo_url</code>
• <code>git push</code>	Upload locally saved commits to GitHub
• <code>git push -u origin</code>	Push a new branch for the first time and link it to the remote <code>git push -u origin branch_name</code>
• <code>git pull</code>	Download the latest updates from GitHub and merge them automatically into your copy
• <code>git fetch</code>	Download updates from GitHub only, without merging them automatically
• Fork	Copy someone else's repository into your own account to work on freely
• Pull Request	A request to merge your changes into the original owner's repository
• README.md	A file describing the project, shown automatically on the repository's main page
• .gitignore	A file listing which files Git should ignore and never push to GitHub

6. Full Development Workflow

From an empty folder to the final push

1	<code>mkdir</code>	Create a new folder for the project
2	<code>cd</code>	Enter the project folder
3	<code>git init</code>	Initialize the Git repository
4	<code>python -m venv venv</code>	Create a virtual environment
5	<code>source venv/bin/activate</code>	Activate the virtual environment
6	<code>pip install</code>	Install the required libraries
7	—	Write the code in your editor
8	<code>git status</code>	Check the current changes
9	<code>git add .</code>	Stage the changes to be saved
10	<code>git commit -m</code>	Save a snapshot of the changes
11	<code>git push</code>	Push the changes to GitHub
12	<code>git pull</code>	Pull down any new updates from GitHub